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Störmer problem restricted to a spherical surface and the Euler and Lagrange tops

Eduardo Piña and Emilio Cortés¹

Departamento de Física, Universidad Autónoma Metropolitana Iztapalapa, PO Box 55-534, Ciudad de México 09340, Mexico

E-mail: pge@xanum.uam.mx and emilco@xanum.uam.mx

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Abstract

In a recent work, Cortés and Poza (2015 *Eur. J. Phys.* **36** 055009) analysed, in full, the dynamics of a charged particle in the field of a magnetic dipole restricted to a spherical surface with the dipole at its centre. This model can be considered as the classical non-relativistic Störmer problem on a sphere. Here, we started from a Lagrangian approach: we derived the Hamilton equations of motion and observed that in this restricted case the equations can be reduced to quadratures, and they were integrated numerically. From the Hamiltonian function we found, for the polar angle, an equivalent one-dimensional system of a particle in the presence of an effective potential. In the present work we start from a change of variable to the cosine of the polar angle. In terms of this variable we obtain an equation that turns out to be the same as the one of a particle in a quartic potential. Then, we can actually solve the equations of motion for the polar angle using Jacobi elliptic functions, and for the azimuthal angle we use the same integrals which were expressed by Jacobi in terms of theta functions, both in the Euler and Lagrange tops. In this restricted Störmer problem, the student at undergraduate or graduate level will have a good example of an integrable nonlinear physical system in which, after analysis of its complex dynamics, one can obtain an analytical solution by means of some special functions of mathematical physics. Additionally, one discovers that the equations of motion of this restricted case of a charge in a magnetic dipole field have the same mathematical structure as the corresponding equations of other well known integrable classical dynamical systems.

Keywords: Störmer problem, charge in a dipole field, Lagrangian and Hamiltonian dynamics, Jacobi elliptic functions, Euler and Lagrange tops

(Some figures may appear in colour only in the online journal)

¹ Author to whom any correspondence should be addressed.

1. Introduction

In a recent work [1], Cortés (with David Cortés Poza), studied the restricted case of the classical problem of a charge constrained to a spherical surface, interacting with the magnetic dipole at the centre of the sphere. The model corresponds to a restricted Störmer problem and it does not represent the general case. However, we believe that, on the one hand, this approach may illustrate the complex dynamics of a point charge in the field of a magnetic dipole and, on the other hand, it could provide some insight into the motion in the general case and thus it can be applied as a simplification to the study of the cosmic radiation and the Earth's field. An early review of the non-restricted problem can be seen in [2]. Later it has been proved that the general problem is not integrable [3], and it has been shown that it is actually chaotic [3]–[6]. However, in this restricted two-dimensional system the equations of motion are integrable in its polar and azimuthal angular variables and for the polar angle we were able to find a one-dimensional equivalent system with an effective potential function [1]. In the present work we establish a dynamical system; we set up the Lagrangian and Hamiltonian functions. From the total energy of the system, and by taking into account the constants of motion, we write a first-order nonlinear differential equation for the cosine of the polar angle, and this happens to be the same differential equation as that of a particle in a quartic potential, which is solved [7] in terms of Jacobi elliptic functions. Then, we can express the analytical solution for the azimuthal angle in terms of the same mathematical integrals which give the Euler angles in terms of elliptic integrals of the third kind either in the Euler case of an asymmetric rigid body far from any torque, or in the symmetric Lagrange top in the field of uniform gravity [8]. We compare some graphs obtained from these analytical results with those obtained in the previous work, where numerical integration was utilized.

We want to point out some elements of this work which should be interesting for graduate (and particularly final-year undergraduate) physics courses. (1) We reduce the set of physical constants to two essential independent parameters. (2) We find that the sign of a parameter related to a root of the potential function determines two different families of orbits, which are oscillatory in the polar angle. There is a bifurcation zero value of this root where we have a non-periodic behaviour similar to the separatrix of several physical problems. (3) The analytical solution we find here allows us to know the expressions for the different periods of the angular variables in terms of our independent parameters; thus by adjusting the relation between periods we are able to compute strictly periodic cases with closed orbits.

This study involves two different fields of physics at the undergraduate or graduate level. First, we have the Lagrangian formulation of the electromagnetic phenomenon and then we have the classical dynamics system that results from this. Then, we can solve the equations of motion for this nonlinear problem by means of some special functions of mathematical physics; this subject belongs to the undergraduate level of physics.

The work is organized in the following way: in section 2 we establish the problem through the Lagrangian approach and we identify the symmetries and the two constants of motion. Then, we derive the Hamilton equations and show that they are integrable.

In section 3 we analyse the dynamics in the polar-angle variable and, by introducing a change to dimensionless variables and parameters, we find the equivalence of the system with a one-dimensional problem with an effective symmetric quartic potential. By taking into account that the energy of the system is an integral of motion, we obtain a first-order differential equation, which represents the energy of a particle of unit mass, with zero value for its total energy, under that quartic potential. In section 4 we analyse the properties and the

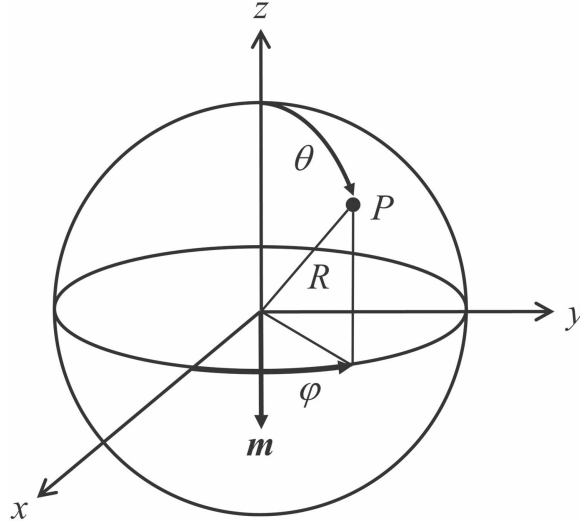


Figure 1. Spherical surface where the charge is constrained. The magnetic dipole is located at the centre of the sphere, with its dipole moment pointing along the negative z -axis. We will use (R, θ, φ) as the spherical coordinates of a point P .

different regimes of this potential. In section 5 we obtain the analytical solution for the motion of the polar and azimuthal angles by using the Jacobi approach to integrate the dynamics of the Euler and Lagrange tops. The new analytical solution is expressed in terms of two dimensionless parameters and a dimensionless time. In section 6 we compute several periodic motions and in section 7 we draw some conclusions.

2. Dynamical system

According to [1], we need to briefly recall our problem formulation and the dynamical equations that were derived, from which we will start the present analysis. We use spherical coordinates, R, θ, φ , to represent the position of the particle at the point P , we set R constant, and we just have two coordinates: the polar and azimuthal angles (see figure 1). We set the magnetic dipole in the centre of the sphere with its magnetic dipole moment $\mathbf{m}$ pointing downwards and M is the mass of the particle. The magnetic vector potential is then expressed as

$$\mathbf{A} = \frac{-\mu_0 m \sin \theta \mathbf{a}_\varphi}{4\pi R^2}, \quad (1)$$

where μ_0 is the magnetic permeability of vacuum and $\mathbf{a}_\varphi$ is a unit vector in the azimuthal direction.

The Lagrangian of the system is written as

$$L = K - U, \quad (2)$$

where K is the kinetic energy of the particle and U is a velocity-dependent potential [9]

$$U = -q \mathbf{v} \cdot \mathbf{A}, \quad (3)$$

where q is the charge of the particle and $\mathbf{v}$ is the velocity vector,

$$\mathbf{v} = R(\dot{\theta} \mathbf{a}_\theta + \sin \theta \dot{\varphi} \mathbf{a}_\varphi), \quad (4)$$

where we use the unit vectors $\mathbf{a}_\theta$ and $\mathbf{a}_\varphi$.

Then, we have

$$U = k\dot{\varphi} \sin^2 \theta, \quad (5)$$

where we define

$$k = \frac{q \mu_0 m}{4\pi R}, \quad (6)$$

and we notice that this parameter k has units of angular momentum.

From equation (4), the kinetic energy is written as

$$K = \frac{1}{2}MR^2(\dot{\theta}^2 + \dot{\varphi}^2 \sin^2 \theta), \quad (7)$$

and the Lagrangian function (2) becomes

$$L = \frac{1}{2}MR^2(\dot{\theta}^2 + \dot{\varphi}^2 \sin^2 \theta) - k\dot{\varphi} \sin^2 \theta. \quad (8)$$

The canonical conjugate momenta are

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = MR^2\dot{\theta}, \quad (9)$$

and

$$p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = [MR^2\dot{\varphi} - k] \sin^2 \theta. \quad (10)$$

Since φ is an ignorable coordinate, due to the symmetry, then p_φ , the conjugated momentum associated with this angle, is a constant of motion.

The Hamiltonian of the system is then

$$H = \dot{\theta} p_\theta + \dot{\varphi} p_\varphi - L = \frac{1}{2}MR^2(\dot{\theta}^2 + \dot{\varphi}^2 \sin^2 \theta), \quad (11)$$

and we find that H is just the kinetic energy of the particle. As we know, the Hamiltonian here is another conserved quantity because L does not depend explicitly on time.

We identify two symmetries in the system, which means that the Lagrangian is invariant under translations in the time t and in the azimuthal angle φ (see Noether's theorem [10]). These symmetries correspond to the two constants of motion: the Hamiltonian or kinetic energy and the generalized momentum p_φ . Now, we can write the Hamiltonian again from equations (9) and (10) in terms of the coordinates and momenta and obtain

$$H = \frac{1}{2MR^2} \left[p_\theta^2 + \frac{(p_\varphi + k \sin^2 \theta)^2}{\sin^2 \theta} \right]. \quad (12)$$

From this function, we write the Hamilton equations:

$$\dot{\theta} = \frac{\partial H}{\partial p_\theta} = \frac{p_\theta}{MR^2}, \quad (13)$$

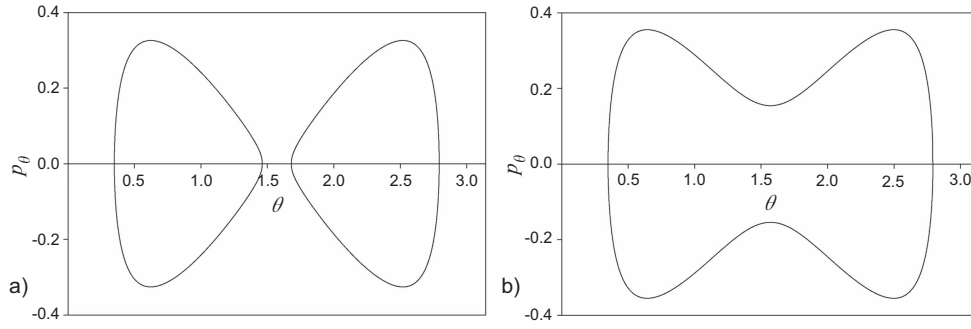


Figure 2. The phase-space diagram of p_θ versus θ for two different values for the constant p_φ . In (a) $p_\varphi = 0.34 k$ and in (b) $p_\varphi = 0.36 k$.

$$\dot{\varphi} = \frac{\partial H}{\partial p_\varphi} = \frac{p_\varphi + k \sin^2 \theta}{MR^2 \sin^2 \theta}. \quad (14)$$

$$\dot{p}_\theta = -\frac{\partial H}{\partial \theta} = \frac{1}{MR^2} (p_\varphi^2 - k^2 \sin^4 \theta) \frac{\cot \theta}{\sin^2 \theta}, \quad (15)$$

$$\dot{p}_\varphi = -\frac{\partial H}{\partial \varphi} = 0. \quad (16)$$

3. The equivalent quartic potential

From the total energy expression equation (12), for the θ coordinate we can write the phase-space expression

$$p_\theta(\theta) = \pm \left[2MR^2 K - \frac{(p_\varphi + k \sin^2 \theta)^2}{\sin^2 \theta} \right]^{1/2}. \quad (17)$$

In figure 2 we plot this function from 0 to π , with, $R = 10$ and $k = 0.5$, with an energy value $K = 1.2155 \cdot 10^{-3}$ and for the constant p_φ we chose two values: $p_\varphi = 0.34 k$, for which the curve exhibits two separated islands, and $p_\varphi = 0.36 k$ for which one obtains a simple closed curve; and all the values are in arbitrary units. We clearly observe a symmetry with respect to the equator, $\theta = \frac{\pi}{2}$. This curve is similar to what is obtained in a ‘bistable potential’, whose phase-space diagram can consist of one single closed curve or two closed curves, depending on the energy value. We point out here that the projection of the particle trajectory on the (θ, p_θ) plane is always a closed path. The path in the spherical surface generally will be non-closed due to the azimuthal motion. In the three-dimensional space $(\theta, p_\theta, \varphi)$ the plot of the trajectory will have a helicoidal shape which lies on the surface of a torus.

From the Hamiltonian, equation (12), where we use K for the kinetic energy or total energy, we have the differential equation for the polar angle

$$2MR^2 K = M^2 R^4 \dot{\theta}^2 + \frac{(p_\varphi + k \sin^2 \theta)^2}{\sin^2 \theta}. \quad (18)$$

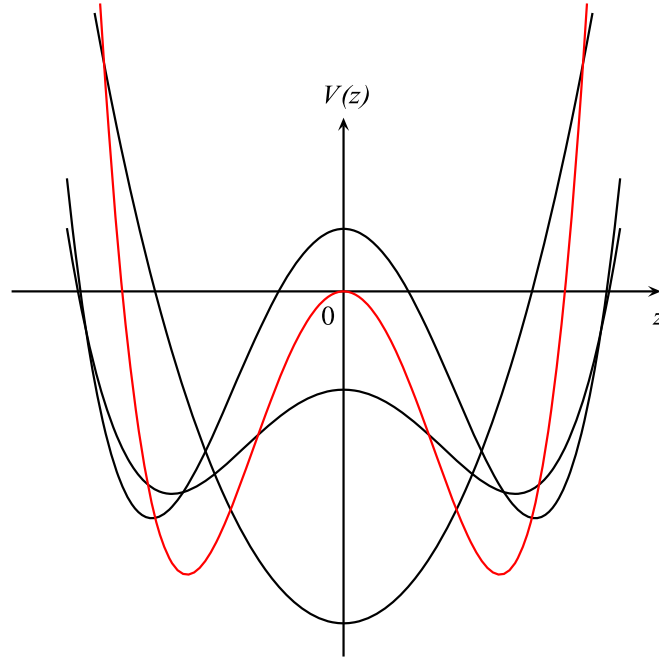


Figure 3. Potential $V(z)$ for four different cases. The case with the maximum at the origin corresponds to a non-periodic case, with three roots; it was computed with $a = -1.5$ and $b = 2.5$. The other cases with inflection points coincide with values of the figures 8(b) and (c) of [1], with values $a = -0.452$, $b = 1.507$, trajectory in one hemisphere (four roots), and values $a = -0.378$, $b = 1.26$, trajectory in two hemispheres (two roots). The third case without inflection points for values $a = -1$, $b = 0.5$, trajectory in two hemispheres (two roots).

Rescaling the time variable to a dimensionless quantity we write

$$\tau = t\sqrt{2K/(MR^2)}. \quad (19)$$

We define the dimensionless parameters:

$$a = p_\varphi / \sqrt{2MR^2K}, \quad b = k / \sqrt{2MR^2K}, \quad (20)$$

where b is assumed to be positive, and we obtain

$$1 = (d\theta/d\tau)^2 + \left(\frac{a + b \sin^2 \theta}{\sin \theta} \right)^2. \quad (21)$$

Now, making a change to a variable

$$z = \cos \theta, \quad (22)$$

and identifying $\dot{z}$ as $dz/d\tau$ we can write

$$\dot{z}^2 + b^2 z^4 + [1 - 2b(a + b)] z^2 + (a + b)^2 - 1 = 0. \quad (23)$$

This equation represents the energy of a particle of unit mass, with zero value for its total energy, where the potential energy has the form

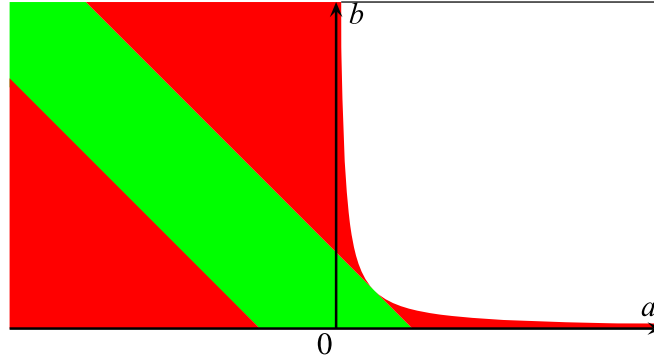


Figure 4. The regions in the space of parameters a and b . We recall that $b > 0$. The white region inside the equilateral hyperbola $4ab = 1$ is the forbidden region, where the potential has complex roots and there is no motion. The light (green) band of the centre is the set of values where the motion is in two hemispheres. The dark (red) regions between the central band and the equilateral hyperbola occur when the motion is in one hemisphere. The separatrix is the parallel lines bounding the two regions of motion.

$$V(z) = (1/2)[b^2 z^4 + (1 - 2b(a + b))z^2 + (a + b)^2 - 1]. \quad (24)$$

The last expression can be factorized in the form:

$$2V(z) = b^2(z^2 - A)(z^2 - B), \quad (25)$$

where

$$A = \frac{2b(a + b) - 1 - \sqrt{1 - 4ab}}{2b^2}, \quad (26)$$

and

$$B = \frac{2b(a + b) - 1 + \sqrt{1 - 4ab}}{2b^2}, \quad (27)$$

which means equation (23) can be written as

$$\dot{z}^2 = b^2(B - z^2)(z^2 - A). \quad (28)$$

4. Properties of the potential function

The function $V(z)$ has the following properties.

- (i) Symmetry with respect to the vertical axis: $V(z) = V(-z)$. As was expected, we have a symmetric double-well potential (or possibly one well). See figure 3.
- (ii) From its definition, the domain of z is inside the interval $(-1, 1)$, since the polar-angle domain is $0 < \theta < \pi$.
- (iii) It is positive at $z = 1, -1$.
- (iv) Note that $B - A = \sqrt{1 - 4ab}/b^2$. For real values of A and B we must have

$$4ab \leq 1, \quad (29)$$

and therefore $B \geq A$. For the existence of at least two real roots of the polynomial $V(z)$, one of these two parameters has to be positive; then it must satisfy the condition $B > 0$. The motion is possible if the potential has at least a real root. We will have a different number of real roots of $V(z)$ according to the sign of A .

- (v) If $4ab = 1$ then $A = B > 0$: there are two double roots in $z = \pm\sqrt{B}$. The particle remains in a horizontal trajectory in any of the two symmetrical minima of the potential. The potential curve only touches the z -axis at these two points. We recall here that the energy has the constant value of zero (see equation (23)).

It will be useful to write the potential energy at $z = 0$; we have from (25)

$$V_0 \equiv V(0) = (b^2/2)AB. \quad (30)$$

There are three cases depending on the sign of A .

- If $A > 0$ then $V_0 > 0$ and we have four real roots symmetric by pairs and the particle will stay trapped in one of the wells. Then, $z = \cos \theta$ will be trapped in the interval $[\sqrt{B}, \sqrt{A}]$ or in the interval $[-\sqrt{A}, -\sqrt{B}]$. In terms of the parameters a and b , this case requires $(a + b)^2 > 1$, the dark (red) region in figure 4.
- If $A < 0$ then $V_0 < 0$ and we have two real roots and the particle goes from one hemisphere to the other. The charge will be trapped in the interval $[-\sqrt{B}, \sqrt{B}]$. In terms of the parameters a and b , this case requires $(a + b)^2 < 1$, the light (green) region in figure 4.
- If $A = 0$, we have two possibilities: the charge is at rest in $z = 0$ at an unstable equilibrium or the charge is trapped on one hemisphere in the semi-closed interval $z \neq 0$ with a non-periodic motion that asymptotically approaches $z = 0$ in an infinite time, for positive or negative values of z . In terms of the parameters a and b , this case requires $(a + b)^2 = 1$ which are the parallel lines separating the light (green) and dark (red) regions in figure 4.

5. Analytical solution in terms of elliptic functions

We consider two different regimes for our potential function: $A > 0$ and $A < 0$.

5.1. The particle trapped in one well (one hemisphere)

In this regime $B > A > 0$. We propose the solution to equation (28) to be

$$z = \cos \theta = \sqrt{B} \operatorname{dn}(u, \kappa), \quad (31)$$

where $\operatorname{dn}(u, \kappa)$ is one of the Jacobi elliptic functions.

To prove it is the solution of that equation, we take the derivative and substitute on the left-hand side to obtain

$$\left(\sqrt{B} \frac{d \operatorname{dn}(u, \kappa)}{du} \dot{u} \right)^2 = (\sqrt{B} \kappa^2 \operatorname{cn}(u, \kappa) \operatorname{sn}(u, \kappa) \dot{u})^2. \quad (32)$$

We substitute on the right-hand side of equation (28) and obtain

$$b^2(B - z^2)(z^2 - A) = b^2 B \kappa^2 \operatorname{sn}^2(u, \kappa) ((B - A - B \kappa^2) + B \kappa^2 \operatorname{cn}^2(u, \kappa)). \quad (33)$$

In the last two expressions we are using some identities of the Jacobi elliptic functions. We equate both expressions and obtain the system of equations

$$B - A - B\kappa^2 = 0, \quad \dot{u}^2 = b^2B, \quad (34)$$

which give us

$$\kappa^2 = \frac{B - A}{B} \leq 1, \quad \dot{u} = b\sqrt{B}, \quad (35)$$

so then one has the analytical solution

$$\cos \theta = \sqrt{B} \operatorname{dn}(b\sqrt{B}\tau, \sqrt{(B - A)/B}). \quad (36)$$

For the azimuthal angle we take equation (14) in terms of the dimensionless time and parameters defined in equations (19) and (20); we have

$$\frac{d\varphi}{d\tau} = \frac{a}{\sin^2(\theta)} + b. \quad (37)$$

Substituting in the analytical solution for $z = \cos \theta$, we write the integral for the angle φ

$$\frac{d\varphi}{d(b\sqrt{B}\tau)} = \frac{1}{\sqrt{B}} + \frac{\frac{a}{b\sqrt{B}}}{1 - B\operatorname{dn}^2(b\sqrt{B}\tau, \kappa)}. \quad (38)$$

This integral can be written mathematically in identical form to the solutions found by Jacobi when he integrates the equations for the asymmetric top without torques, the Euler top, and the symmetric top with uniform gravity, which is the Lagrange top.

In the case of the Euler top, in terms of the Euler angles, the unit vector directed along the angular-momentum vector has the components in spherical coordinates of the inertial frame

$$(\sin \theta \sin \psi, \sin \theta \cos \psi, \cos \theta), \quad (39)$$

where ψ and θ are Euler angles. We use the same notation θ for both angles since we are going to identify them.

In the Euler's problem we can write $\cos \theta$ in two forms, which will be used in this paper. One of them is

$$\cos \theta = \operatorname{sn}(\alpha, \kappa') \operatorname{dn}(u, \kappa), \quad (40)$$

where $\kappa^2 + \kappa'^2 = 1$, u is linear in time and α is a constant. We see that $\cos \theta$ of both our problem and Euler's coincide if we assume

$$\sqrt{B} = \operatorname{sn}(\alpha, \kappa'). \quad (41)$$

Then, surprisingly, the third Euler angle ϕ , and our angle φ differ the most for an angle with constant angular velocity

$$\frac{d\varphi}{d(b\sqrt{B}\tau)} = \frac{1}{\sqrt{B}} + \frac{\frac{\operatorname{cn}(\alpha, \kappa') \operatorname{dn}(\alpha, \kappa')}{\operatorname{sn}(\alpha, \kappa')}}{1 - \operatorname{sn}^2(\alpha, \kappa') \operatorname{dn}^2(b\sqrt{B}\tau, \kappa)}, \quad (42)$$

where we have replaced, besides equation (41),

$$\frac{a}{b\sqrt{B}} = \frac{\operatorname{cn}(\alpha, \kappa') \operatorname{dn}(\alpha, \kappa')}{\operatorname{sn}(\alpha, \kappa')}, \quad (43)$$

which are derived from identities of elliptic functions. We have computed

$$\operatorname{cn}(\alpha, \kappa') = \sqrt{1 - B}, \quad \operatorname{dn}(\alpha, \kappa') = \sqrt{1 - A} \quad (44)$$

as well as the property

$$(1 - A)(1 - B) = \frac{a^2}{b^2}. \quad (45)$$

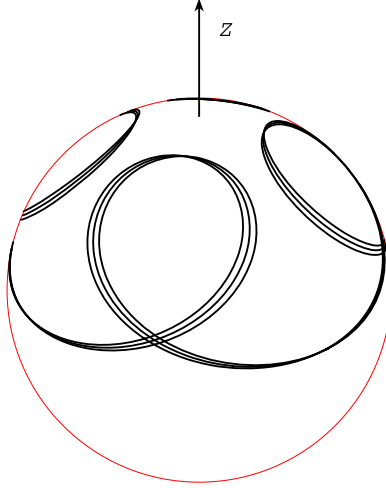


Figure 5. The motion of the charge in one hemisphere. Values of parameters coincide with figure 8(b) in [1].

The same integral appears in the sum of the Euler angles $\phi + \psi$ in the motion of the symmetric Lagrange top, with a characteristic definition for the constant parameter α , although in the Lagrange case the Euler θ angle and ours are very different.

The second integral in equation (42) is the same in the three cases.

As an example of this regime, we computed (using the analytical solution) the case represented in paper [1] by figure 8(b), as shown in figure 5.

5.2. The particle goes from one hemisphere to the other

Here, we have $A < 0$ and $B > 0$; then we propose the solution to equation (28):

$$z = \cos \theta = \sqrt{B} \operatorname{cn}(u, \kappa). \quad (46)$$

To prove it is the solution of that equation we take the time derivative and substitute on the left-hand side of that equation and obtain

$$\dot{z}^2 = B \operatorname{sn}^2(u, \kappa) \operatorname{dn}^2(u, \kappa) \dot{u}^2 = B \operatorname{sn}^2(u, \kappa) [1 - \kappa^2(1 - \operatorname{cn}^2(u, \kappa))] \dot{u}^2. \quad (47)$$

Substituting on the right-hand side of the same equation

$$\kappa^2 \dot{u}^2 = b^2 (B - z^2)(z^2 - A) = b^2 B \operatorname{sn}^2(u, \kappa) (B \operatorname{cn}^2(u, \kappa) - A). \quad (48)$$

In the last two expressions we have used some identities of the Jacobi elliptic functions.

We equate both expressions and obtain the system of equations

$$(1 - \kappa^2) \dot{u}^2 = -b^2 A, \quad \kappa^2 \dot{u}^2 = b^2 B. \quad (49)$$

We solve this system to deduce, in this case,

$$\kappa^2 = \frac{B}{B - A} \leq 1, \quad \dot{u} = b \sqrt{B - A}, \quad (50)$$

from which we have the analytical solution

$$\cos \theta = \sqrt{B} \operatorname{cn}(b \sqrt{B - A} \tau, \sqrt{B/(B - A)}). \quad (51)$$

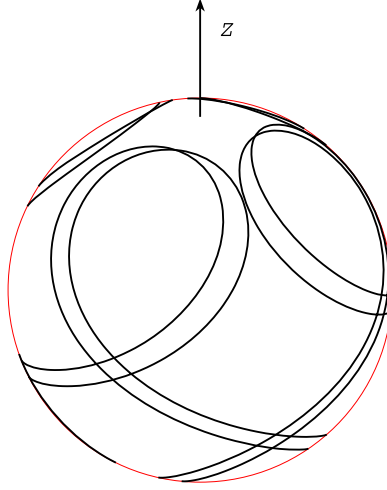


Figure 6. The motion of the charge in two hemispheres. Values of parameters coincide with figure 8(c) in [1].

In this case, equation (37) for the azimuthal angle turns into

$$\frac{d\varphi}{d(b\sqrt{B-A}\tau)} = \frac{1}{\sqrt{B-A}} + \frac{\frac{a}{b\sqrt{B-A}}}{1 - B \operatorname{cn}^2(b\sqrt{B-A}\tau, \kappa)}. \quad (52)$$

Here, this integral can also be written mathematically in an identical form to the solutions found by Jacobi when he integrates the equations for the asymmetric top without torques: the Euler top. As stated before, there is another case in the motion of the Euler top in which $\cos \theta$, both for the Euler θ angle and for our angle, is written as

$$\cos \theta = \operatorname{cn}(\beta, \kappa') \operatorname{cn}(u, \kappa), \quad (53)$$

where $\kappa^2 + \kappa'^2 = 1$, u is linear in time and β is a constant. In order to have an equivalence we need, for our case,

$$\sqrt{B} = \operatorname{cn}(\beta, \kappa'). \quad (54)$$

Then, surprisingly, the Euler angle ϕ for the Euler top, and our angle φ differ the most for an angle with constant angular velocity:

$$\frac{d\varphi}{d(b\sqrt{B-A}\tau)} = \frac{1}{\sqrt{B-A}} + \frac{\frac{\operatorname{sn}(\alpha, \kappa') \operatorname{dn}(\alpha, \kappa')}{\operatorname{cn}(\alpha, \kappa')}}{1 - \operatorname{cn}^2(\alpha, \kappa') \operatorname{cn}^2(b\sqrt{B-A}\tau, \kappa)}, \quad (55)$$

where we have replaced, in addition to equation (54),

$$\frac{a}{b\sqrt{B-A}} = \frac{\operatorname{sn}(\alpha, \kappa') \operatorname{dn}(\alpha, \kappa')}{\operatorname{cn}(\alpha, \kappa')}, \quad (56)$$

which are derived from identities of elliptic functions. We have used the properties

$$\operatorname{sn}(\beta, \kappa') = \sqrt{1-B}, \quad \operatorname{dn}(\beta, \kappa') = \sqrt{\frac{B(1-A)}{B-A}}, \quad (57)$$

as well as the relation (45).

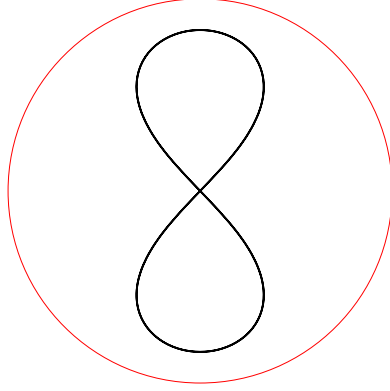


Figure 7. Periodic motion of the charge in two hemispheres. Here, there is an oscillatory behaviour in both: the polar and the azimuthal angles. We found the same figure in the ‘eight’ form as in [1]. The observer is in the plane of the equator.

The second integral in equation (55) actually has the same formal expression as the second integral in the previous case, equation (42), since one can be transformed in the other by the parameter transformation

$$\beta = K(\kappa') - \alpha, \quad (58)$$

where $K(\kappa)$ is $\frac{1}{4}$ of the real period of the functions $\text{sn}(u, \kappa)$ and $\text{cn}(u, \kappa)$.

As an example of this regime we computed the case represented in paper [1] by figure 8(c) using the analytical solution, as we show in figure 6.

6. Periodic solutions

In paper [1], Cortés and Cortés Poza found a periodic solution with the form of an ‘eight’ over the sphere for the values of the physical constants $M = 2$, $R = 10$, $k = 0.5$, $\theta(0) = \pi/3$, $p_\varphi = -0.722k$. As we see, in this case we have an oscillatory behaviour in both: the polar and the azimuthal angles.

From our explicit expressions in this paper we have found a more exact value of $-0.722\,079\,599\,230\,17$ for $p_\varphi/k = a/b$. The example considered in [1] has been reproduced in figure 7. The observer is in the plane of the equator.

We found these periodic solutions for $a < 0$ when the integral in a period $2K(\kappa)$ of the right-hand side of equation (42) is zero. The possible periodic solutions are on a curve in the space of parameters a and b . Some values on this curve are collected in the following table.

a	b
-0.239 801 671 792 545	1
-0.778 607 957 509 375	1.5
-1.296 926 866 373 04	2
-1.807 581 471 141 74	2.5
-2.314 546 818 025 53	3
-2.819 455 456 856 58	3.5
-3.323 100 840 882 73	4
-3.825 914 897 569 56	4.5
-4.328 152 832 949 38	5
-4.829 975 109 409 12	5.5

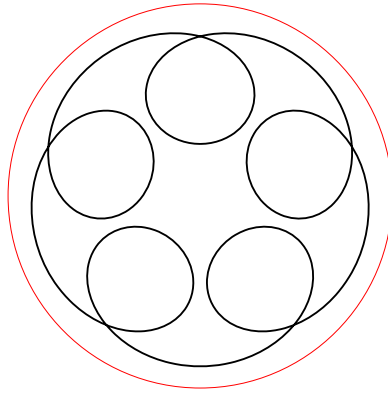


Figure 8. Periodic motions of the charge in one hemisphere. Here, we have a periodic orbit with five loops computed with $a = -1/2$ and $b = 1.589\,501\,739\,525\,29$. The view is from below the south pole.

For values lower than $b = 0.792\,767$ we have not found the corresponding value for a to have a periodic solution; the explanation is that if $a \rightarrow 0$ then $B \rightarrow 1$.

Other periodic solutions are found where the azimuthal angle rotates uniformly and the integral over an entire number of periods $2K(\kappa)$ of the azimuthal angle is a rational multiple of 2π . An example for period 5 is found with the values of parameters $a = -0.5$, $b = 1.589\,501\,739\,525\,29$ as in figure 8.

7. Conclusions

In this work we present the analytical solution of the equations of motion derived in a recent publication [1], where the dynamics of a charged particle in the field of a magnetic dipole restricted to a spherical surface with the dipole at its centre are analysed in full detail. Here, the authors started from a Lagrangian approach: the Hamilton equations of motion were derived and it was observed that in this restricted case the equations can be reduced to a quadrature and the equations of motion were integrated numerically. From the Hamiltonian function, for the polar angle, an equivalent one-dimensional system of a particle in the presence of an effective potential has been found.

In the present work we started from a first-order differential equation for the cosine of the polar angle, which is the same as that of a particle in a quartic potential.

We studied the properties of this potential function and established different regimes which depend on just two independent parameters related to the two constants of motion. For the polar angle, essentially two different oscillatory regimes which determine a one-hemisphere motion or two-hemisphere motion for the dynamics of the particle are found. Between these two regimes we have a parameter bifurcation value where there is a non-periodic behaviour similar to the separatrix of several physical problems. For each regime we solved the differential equation (28) in terms of Jacobi elliptic functions. We have explicitly written the analytical solution for the polar angle, and from this we found that the analytical solution for the azimuthal angle of this restricted Störmer problem can be expressed in terms of the same mathematical integrals which give the Euler angles in terms of elliptic integrals of the

third kind. This applies to both the Euler case of an asymmetric rigid body far from any torque, or for the Lagrange top in the field of uniform gravity.

Finally, we point out that we have compared graphs obtained from these analytical results and they are in agreement with those obtained previously from a numerical integration and, furthermore, the analytical solution we found here allows us to know the expressions for two different periods in terms of our independent parameters. Thus, we were able to compute strictly periodic cases with closed orbits.

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